

Full stack developer proposed roadmap

The full stack developer is expected to contribute massively in the migration of the old website to the new website (Actual TOR during the call was very specific on this) and work alongside priority tasks on Climsa activities (climate hub system, the rcc website and other climate products that require harmonisation he will also support the multi-hazard platform in data ingestion during the duration.

Overall Roadmap

Period	Dates	Main Focus
Month 1	Aug 3 - Sep 2, 2026	ACMAD weather/climate product migration into ClimWeb
Month 2	Sep 3 - Oct 2, 2026	RCC website revamp and RCC activity/product integration
Month 3	Oct 5 - Oct 30, 2026	ClimateHub enhancement, CLIMSA website, final alignment and handover

1. Website migration: to be done in two phases

a. Phase one main tasks

i. Product Audit and Mapping

- List ACMAD weather products: Multi-Hazard Outlook, daily rainfall observation, satellite rainfall, ITD/ITCZ, RDT, GFS meteograms, ensemble rainfall forecasts, heat index, weekly forecasts, etc.
- List climate products
- Decide which products become the websites `ProductPage`, product item, map layer, static bulletin, or external app link.
- **Output:** product migration matrix.

ii. Product Structure Alignment

- Define product categories: `Weather`, `Climate`, `Drought`, `Multi-Hazard`, `Forecasts`, `Monitoring`.

- Align product metadata: frequency, validity, issue date, archive behavior, thumbnail, files, map layer availability
- Configure homepage featured products and Multi-Hazard Overview .
- Output: final content/data model pla

iii. THREDDS Dataset Inventory

- Identify THREDDS catalogs
- Document variables, dimensions, time conventions, bounding boxes, units, update frequency.
- Output: THREDDS dataset registry.

iv. External Data Puller/API Script Design (The website shall have its own endpoint that allows data to be ingested but the script that pushes the data shall be external to the system and can run on premise or the cloud)

- Build an external script/service that fetches datasets from THREDDS.
- The script can run outside website system via cron, Celery beat, Airflow, or systemd timer.
- Normalize outputs into agreed formats: NetCDF, GeoTIFF/COG, PNG preview, JSON metadata.
- Push or expose results to ClimWeb through API endpoints or shared storage (minio cloud instance).
- Output: ingestion architecture and script specification.

v. Website API Layer

- Create APIs for:
 - product catalog
- Output: API endpoints usable by the system frontend and external scripts.

vi. Content Migration

- Move descriptions, titles, thumbnails, and product explanations from ACMAD into Website.
- Clean naming and remove duplicated/legacy labels.
- Ensure Latest Updates is used instead of Latest Statements .
- Output: populated Web pages.

vii. Testing and Validation

- Validate THREDDS pulls.
- Validate product freshness and archive ordering.
- Test API responses.

- Test mobile/desktop UI.
- Output: QA report and fixes.

viii. Deployment, Documentation, and Handover (The system is already deployed with a CI/CD pipeline hence there will be little to do other than documentation and eventually domain migration)

- Document product setup.
- Document external script configuration.
- Document THREDDS dataset registry, and how experts can push data directly to the system due to THREDDs capacity and internet challenges
- Train editors on adding/updating products.
- Output: deployment checklist and handover guide

Phase1	Dates	Tasks
Formalisation of way of working code repository, rules of making changes to code bases and tools to use	3 Aug 2026	Added to github set up the repository on the computer and tests out that it works, makes some changes on the code push, makes peer review of the code and walk through the entire process to deployment once changes are merged
Product selection & mapping	Aug 3 - Aug 4	Select 6-8 priority weather/climate products, define categories, metadata, update frequency
THREDDS dataset inventory	Aug 4 - Aug 6	Confirm endpoints, variables, dimensions, formats, access method, sample files
External fetch script/API design	Aug 6 - Aug 10	Build external pull script pattern, normalize outputs, create metadata JSON, define API contract

ClimWeb product/API alignment	Aug 10 - Aug 13	Align product pages, latest/archive structures, product listing APIs, map layer metadata
UI alignment	Aug 13 - Aug 16	Homepage Multi-Hazard Overview , Weather/Climate product sections, Featured Products, Latest Updates
Content migration	Aug 14 - Aug 17	Move initial descriptions, thumbnails, product summaries, latest issue references
QA, fixes & handover	Aug 17 - Aug 19	Test THREDDS pulls, API responses, map preview, responsive UI, document setup

Deliverables by Aug 19

- 6-8 migrated priority products.
- External THREDDS pull script pattern.
- Basic product/API integration.
- Homepage **Multi-Hazard Overview** .
- Weather and Climate product groupings.
- Product listing/detail pages aligned to ACMAD content.
- Initial documentation and handover notes.

Phase 2 shall entail:

- Full historical archives.
- All ACMAD products.
- Advanced product-specific visualisation.
- Automated monitoring/retry system.

Phase 2 Tasks

Dates	Tasks
Aug 20 - Aug 21	Add remaining high-priority ACMAD products not covered in Phase 1
Aug 21 - Aug 24	Improve archive handling: date filters, latest item detection, older product browsing
Aug 24 - Aug 26	Add product-specific map/view/download behavior for datasets from THREDDS
Aug 26 - Aug 28	Add validation and error handling for failed pulls, stale datasets, missing metadata
Aug 28 - Aug 31	Improve UI polish for product pages, Multi-Hazard Overview, Featured Products, and Latest Updates
Sep 1 - Sep 2	Final QA, documentation, deployment checklist, editor/admin handover

Month 2: RCC Website Revamp

Dates	Tasks	Deliverables
Sep 3 - Sep 5	RCC content audit and structure planning	RCC sitemap, content inventory, migration matrix
Sep 8 - Sep 12	RCC homepage and core pages design/alignment	RCC homepage, About RCC, Services, Products, Activities structure
Sep 15 - Sep 19	RCC activities integration	ACCOF pages, RCC events, meetings, workshops, activity archive
Sep 22 - Sep 26	RCC products and bulletins integration	Bulletins listing, product categories, latest bulletin blocks, archive logic

Sep 29 - Oct 2	RCC QA and editor workflow	Responsive checks, content review, admin/editor documentation
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RCC Website Key Tasks

- Revamp RCC Africa website using ClimWeb source code, similar to the Summer School customization approach.
- Create RCC-specific navigation and homepage.
- Add RCC Activities section.
- Add RCC Products and Bulletins sections.
- Integrate ACCOF and related RCC events.
- Add archive/listing pages for bulletins, outlooks, reports, and meeting outputs.
- Link RCC products to migrated ClimWeb products where relevant.
- Ensure ACMAD branding remains consistent but RCC has its own clear identity.

Month 3: ClimateHub, CLIMSA, Final Alignment

Dates	Tasks	Deliverables
Oct 5 - Oct 9	ClimateHub enhancement planning and page additions	ClimateHub page plan, improved navigation/content structure
Oct 12 - Oct 16	ClimateHub implementation support	Additional pages, better links to products, RCC, bulletins, map resources
Oct 19 - Oct 23	CLIMSA website customization	CLIMSA homepage, project pages, activities, resources, partners, outputs
Oct 26 - Oct 28	Cross-site alignment	Shared navigation logic, consistent branding, links between ACMAD/RCC/CLIMSA/Climate Hub
Oct 29 - Oct 30	Final QA and handover	Documentation, deployment notes, editor guide, final

ClimateHub Tasks

- Keep ClimateHub as a standalone system.
- Add missing pages such as About, Data Catalog, Products, Tools, User Guide, Partners, and Contact/Support.
- Improve links between ClimateHub datasets and ClimWeb product pages.
- Improve discoverability of map layers and datasets.
- Add landing pages for key dataset groups.
- Add documentation/help content for users.

CLIMSA Website Tasks

- Build or customize CLIMSA website using ClimWeb source code.
- Add CLIMSA homepage.
- Add project overview, objectives, activities, countries/partners, resources, news/events, publications, and outputs.
- Align visual style with ACMAD but keep CLIMSA project identity.
- Add content management fields so editors can update activities and resources.

Priority Order

1. Finish ACMAD product migration.
2. Revamp RCC website.
3. Integrate RCC products, bulletins, and activities.
4. Enhance ClimateHub pages and linking.
5. Build/customize CLIMSA website.
6. Final cross-site polish and handover.

Final Deliverables By Oct 30, 2026

- Migrated ACMAD weather/climate products in ClimWeb.
- Revamped RCC website based on ClimWeb.
- RCC activities, products, and bulletins integrated.
- Enhanced ClimateHub with additional pages and better navigation.
- CLIMSA website customized from ClimWeb.
- Documentation for editors/admins.

- Final QA and deployment checklist.

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Progress review - status as at 21 September 2026

This review compares the original delivery plan with implementation evidence in the ClimWeb repository. The original eight pages are retained unchanged. Status is based on merged code, documentation, tests, migrations, templates, and operational tooling; it does not treat a planned date alone as proof of completion.

Overall Roadmap Progress

Period	Dates	Status	Verified progress
Month 1	Aug 3 - Sep 2, 2026	SUBSTANTIALLY ACHIEVED	ACMAD product migration framework delivered with 20 product import commands, configurable sources, scheduling, monitoring, retry/stop controls, archives, redesigned product pages, featured products, tests, and documentation. Some architecture and handover items remain open.
Month 2	Sep 3 - Oct 2, 2026	IN PROGRESS	RCC homepage, services and climate-product catalogues, hierarchical dataset browsing, events integration, and ARC2, CPC, seasonal-map and EIN15 import/archive workflows are in place. Bulletin coverage, final QA and editor handover are not yet fully evidenced.
Month 3	Oct 5 - Oct 30, 2026	NOT YET DUE	No repository evidence yet for the planned ClimateHub enhancement, CLIMSA customization, cross-site alignment, or final handover.

Completion Snapshot

- Product migration breadth is ahead of the original 6-8 product target: 20 ACMAD import commands and 19 product-import test modules are present.
- RCC delivery is already operational in several areas scheduled for later in Month 2, including public archives and product-specific dataset workflows.
- Important work outside the initial roadmap has also been delivered: OJT, capacity-building participant mapping, event migration, staff profile management, and product subscriptions/notifications.
- The principal gaps are a consolidated THREDDS registry, the originally proposed standalone external pusher/API contract, complete RCC bulletin coverage, editor handover, ClimateHub, and CLIMSA.

Status definitions: Achieved = implementation evidence is present; In progress = meaningful implementation exists but the full roadmap outcome is not evidenced; Not yet due = scheduled after this review date.

Month 1: ACMAD Product Migration Review

Comparison of the original Phase 1 and Phase 2 tasks with verified repository outcomes.

Roadmap item	Status	Evidence-based comparison
Ways of working / repository setup	ACHIEVED	Active GitHub workflow with feature branches, peer-review pull requests, merges, tests, and deployment-oriented changes.
Product audit and mapping	ACHIEVED	Legacy-product audit tooling and a proposed taxonomy are present, with product categories and importer families represented in code and documentation.
Product structure alignment	ACHIEVED	Product index/detail pages, category filtering, metadata, validity, latest-item logic, featured-product ordering, thumbnails, maps, documents, and archive behavior were implemented.
THREDDS dataset inventory	IN PROGRESS	THREDDS discovery and source configuration support multiple products and RCC datasets. A single consolidated registry documenting every variable, unit, bounding box, and update frequency is not yet evidenced.
External puller / API script design	PARTIAL / ADAPTED	Configurable importers, Celery scheduling, normalization, monitoring, retries, stop controls, and source resets are implemented. The original standalone external pusher and formal website ingestion API contract are not clearly evidenced.
Website API layer	PARTIAL	Internal service, task, model, admin, and view layers support ingestion and product delivery. A separately documented public product-catalog API remains unverified.
Content migration	SUBSTANTIALLY ACHIEVED	Twenty ACMAD import commands cover priority weather and climate families; product pages, summaries, latest-file behavior, thumbnails, and historical items are supported.
Testing and validation	ACHIEVED FOR IMPLEMENTED SCOPE	Nineteen product-import test modules plus monitoring, source, model, UI and notification tests are present. Import logs, errors, retries, cancellation, and freshness/latest-item behavior were implemented.
Documentation and handover	IN PROGRESS	Product migration, importer operation, product management, and subscription documentation exist. Formal editor training, complete handover guide, domain migration, and final deployment checklist are not yet evidenced.

Additional Month 1 Delivery

- Product subscriptions and notifications: subscriber preferences, confirmation/unsubscribe flows, HTML email templates, import-triggered notifications, dashboard controls, CSV export, pagination, and documentation.
- Homepage and product UX: featured-product ordering, configurable hero content, a Multi-Hazard map presentation, Latest Updates improvements, responsive layouts, and configurable theme/content components.

Month 2: RCC Website Revamp Review

Status reflects delivery through 21 September 2026. Tasks scheduled after that date are not marked late.

Roadmap item	Status	Verified progress
RCC content audit and structure planning	SUBSTANTIALLY ACHIEVED	RCC-specific homepage, navigation direction, services structure, product catalogue, dataset catalogue, and activities/event presentation are represented in the implementation.
RCC homepage and core pages	ACHIEVED	RCC landing page, refined homepage layout and map, partners, services catalogue, climate-products catalogue, and branded presentation are implemented.
RCC activities integration	SUBSTANTIALLY ACHIEVED	RCC service pages include event presentation. ACMAD event migration, event-detail redesign, meeting integrations, and Board of Governors pages provide broader activity support.
RCC products and bulletins integration	IN PROGRESS / EARLY DELIVERY	Climate-product and data-service catalogues are present. ARC2, CPC rainfall, seasonal maps, and EIN15 have managed import workflows; seasonal maps and EIN15 include public archives. Complete bulletin categories and all planned bulletin archives are not yet evidenced.
Product-specific map / view / download	ACHIEVED FOR CURRENT DATASETS	Hierarchical dataset browsing, country and station views, list/grid controls, search, public map galleries, previews, and downloads are implemented for the current RCC data scope.
Validation, monitoring and error handling	ACHIEVED FOR CURRENT DATASETS	Importer tests, source validation, run records, logs, visible errors, progress, scheduling, retry/stop behavior, and administrative controls are present across product and RCC workflows.
RCC QA and editor workflow	UPCOMING	Implementation tests and responsive refinements exist, but the roadmap's Sep 29 - Oct 2 content review and editor/admin handover are still ahead of the review date.

Month 3: ClimateHub, CLIMSA and Final Alignment

The planned work starts on 5 October 2026. No implementation evidence was identified yet for ClimateHub page expansion, CLIMSA website customization, cross-site alignment, or final project handover. These items remain the next major roadmap phase.

Additional Achievements To Add To The Roadmap

The following completed work was not explicitly captured in the original roadmap and should now be recognized as delivered scope.

Added achievement	Delivery recorded
On-the-job training (OJT)	Dedicated editable OJT service page, content blocks, deployment seeding, styling, navigation placement, and tests.
Capacity-building participant management	Participant register, Africa choropleth, country summaries, interactive filters, paginated table, gallery modal, admin-managed color scheme, reusable page block, seed data, and tests.
Events migration	ACMAD event import command with image/document downloads, update-existing behavior, safe unpublished defaults, rich-text sanitation and repair tooling, routing/style fixes, and redesigned event detail pages.
Staff profile management	Invitation-only profile onboarding, secure password setup/reset, staff review, account linking, profile editing, employment/offboarding workflow, contact/social links, page selections, documentation, and extensive automated tests.
Product subscriptions and notifications	Configurable subscriptions, granular preferences, import-triggered notifications, HTML emails, subscriber dashboard, CSV export, pagination, duplicate safeguards, and operational documentation.
Meeting and calendar support	Google Meet event integration and Google Calendar OAuth settings were added to support event operations.
Institutional content management	Editable African meteorological services directory, Director General message, Board of Governors pages, dashboard-managed FAQ drawer, and additional homepage/content controls.

Revised Priority Order

	Priority	Required outcome
1.	Close Month 1 gaps	Publish the consolidated THREDDS registry and formal ingestion/API contract; complete editor training and handover documentation.
2.	Complete RCC Month 2	Finish bulletin coverage and archives, validate content, complete responsive QA, and document the editor workflow.
3.	Enhance ClimateHub	Deliver the planned pages, dataset-group landing pages, discoverability improvements, and links to ClimWeb/RCC products.
4.	Build the CLIMSA website	Implement the project homepage, activities, partner/country, resource, publication, news/event, and output sections.
5.	Final alignment and handover	Complete cross-site navigation/branding, deployment checks, editor/admin guidance, and final acceptance review.